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Creating a report using INBOmd

Thierry Onkelinx

Where to find documentation

- Documention on INBOmd
  - https://inbomd.netlify.com
- Example code using INBOmd
  - https://inbomd-examples.netlify.com/
  - Do have a look at our tips and tricks for the pdf version
- bookdown: Authoring Books and Technical Documents with R Markdown
  - https://bookdown.org/yihui/bookdown/
- R Markdown: The Definitive Guide
  - https://bookdown.org/yihui/rmarkdown/

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Install requirements

Install tinytex

```
update.packages(ask = FALSE, checkBuilt = TRUE)
if (length(find.package("tinytex", quiet = TRUE)) == 0) {
  install.packages("tinytex")
  tinytex::install_tinytex()
}
```

- tinytex is only required for pdf output
- tinytex is not compatible with MikTeX, uninstall MikTeX first

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Install INBOmd

```r
if (length(find.package("remotes", quiet = TRUE)) == 0) {
  install.packages("remotes")
}
remotes::install_github("inbo/INBOmd", dependencies = TRUE)
# setup for tinytex
tinytex::tlmgr_install(c(
  'inconsolata', 'times', 'tex', 'helvetic', 'dvips'
))
tinytex::tlmgr_conf(
  c("auxtrees", "add", system.file("local_tex", package = "INBOmd"))
)
tinytex::tlmgr_install(c("hyphen-dutch", "hyphen-french"))
```

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Getting started

Important things to known about bookdown

- There must be at least a file named `index.Rmd`
  - This contains the YAML header
  - It must contain at least one header 1 (#)
  - Pro tip: place your summary or introduction in `index.Rmd`
- You can split long documents over multiple `.Rmd` files
  - Add only a YAML to the `index.Rmd`
  - All `.Rmd` files will be glued into a single `.Rmd` when rendering the document
    - Except files starting with _
    - Their order: `index.Rmd` + other files in alphabethical order
  - Pro tip: start the filenames with a number indicating the order

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Starting a new document

- Open RStudio
- Choose File > New File > R Markdown
- Select 'From Template' and select 'INBO rapport'
- Enter the `name` for your report, select its main directory and press 'OK'
  - This creates `name/name.Rmd` in the selected directory
  - Rename `name/name.Rmd` to `name/index.Rmd` in the `Files` window of RStudio
- Choose File > New Project > Existing directory, select `name` and press 'Create project'
- Open `_bookdown.yml` in this project and update `book_filename` and `output_dir`
- Open `index.Rmd` in this project and update the YAML header

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Coverting an existing RMarkdown document

- Start a new Bookdown document
- Copy-paste the relevant code from the existing document to the new document
- Try this using your own RMarkdown document

The `setup` chunk

- The `setup` chunk is the best location to load packages
- `opts_chunk$set()` defines the default options for each chunk
  - You can override these in an individual chunk
  - `fig.width` and `fig.height` are defined in inches
    - `fig.width = 150 / 25.4` defines the plot width to be 150 mm (5.91 in)
- `theme_set()` and `interactive()`
  - Sets a default INBO theme for ggplot2
  - pdf output uses a smaller font size for the ggplot2 graphics
  - You can use the Flanders Art Sans font if it is installed on your computer

Building the report

- Use the `Build Book` button instead of the `Knit` button
  - You can find this button in the same window as `Environment` and `History`
- Open the dropdown menu to select the required output format
  - Defaults to **all** formats listed in the YAML
  - Pro tip: select a **single** format while working on the document
- Try this with your first Bookdown document

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Figures in bookdown

Figure basics

Chunk setup

- One plot = one chunk
- Give the chunk a relevant name
  - Only alphanumeric characters (a-z, A-Z, 0-9), slashes (/), or dashes (-)
  - Chunk names must be unique

Reference a figure

- Syntax: `\@ref(fig:the-chunk-name)`
- This will render the figure number and a hyperlink to the caption
- Works only if the figure has a caption

Add an external figure

- Create an R chunk
- Add `knitr::include_graphics("path-to-figure.png")`

Add a caption

- Plain text caption without special characters (%, _)
  - Chunk option: `fig.cap = "Place your caption here"`
- Special characters or Markdown syntax
  - Use text references
  - Outside of chunk:
    - `(ref:label) This figure caption _works_ 100%!`
    - Must be a single line
    - At least 1 blank line above and below
  - Chunk option: `fig.cap = "(ref:label)"`
  - `label` must be a unique name in the document
  - Pro tip: use the chunk name as `label`

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Tables in bookdown

Tables

- Chunk setup identical to chunk setup for figures
- First try `knitr::kable()` to generate the table
  - See examples on the INBOmd website
  - Improve formatting with `kableExtra`
- `knitr::kable(caption = "My table caption")`
  - Text references `(ref:label)` are possible too
- Reference the table with `\@ref(tab:the-chunk-name)`

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Citing literature

Setting up the YAML

- Add `bibliography: your-bibliography.bib` to the YAML header
  - Must have `.bib` extention
  - Name contains only letters, digits and dash (-)
- Must be in bibtex format
- Add the code below to add a title for the references in HTML or EPUB (not necessary for PDF)

```
`r if (knitr::is_html_output()) '# References {-}'`
```

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Inserting citations manually

- `[@bibtex-key]` for the notation with parantheses (Agresti, 2002)
- `@bibtex-key` for the notation without parantheses Amano (2012)
- Use a semi-colon to separate citations
  - `[@bibtex-key; @bibtex-key2; @bibtex-key3]` (Anselin *et al.*, 2014; Banerjee *et al.*, 2003)
- Create a sensible key for each reference in your bibliography
  - Key must contain only letters, digits, dashed and colons
  - Edit the bibliography using RStudio or JabRef

Inserting citations using `citr` addin

- `install.packages("citr")`
- Go to “Addins” in the RStudio toolbar
- Select “CITR”, “Insert citations”
- Search and select the references you want
- Insert the citations

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Using Zotero through `citr` addin

- Install “Zotero”” and “Better bibtex” plugin
- Start Zotero
- Go to “Addins” in the RStudio toolbar
- Select “CITR”, “Insert citations”
- Connect to Zotero
- Go to “Settings” tab
- Add references to local bibliography file

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Miscellaneous

Footnotes

- `My sentence with a footnote^[footnote text].`
- My sentence with a footnote[1].
- Use `[^footnote-label]: footnote text` to define the footnote in a separate paragraph[2].
  - `[^example]: This an example of a reusable footnote.`
- These footnotes are reusable[3].
  - Note that reusing a footnote[4] adds a copy of the footnote.

---

[1] footnote text
[2] This an example of a reusable footnote.
[3] This an example of a reusable footnote.
[4] This an example of a reusable footnote.

Equations

- Equations are based on LaTeX code.
- Online editor
- Code between single $ results in an inline equation $\sum_{i=1}^{n} x_i$
- Code between double $ results in a stand alone equation

$$\sum_{i=1}^{n} x_i$$

- Code between `\begin{equation}` and `\end{equation}` and with a label (`\#eq:sum`) is stand alone and numbered

$$\sum_{i=1}^{n} x_i \tag{1}$$

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Cross references

- Figures: `\@ref(fig:chunk-name)`
- Tables: `\@ref(tab:chunk-name)`
- Equations: `\@ref(eq:equation-label)`
- Headings
  - Label the heading by adding `(#my-label)` at the end
  - `\@ref(my-label)`
  - Pro-tip: use a prefix to indicate the heading level
    - `# Introduction (#ch:introduction)`
    - `## Conclusion (#s:executive-conclusion)`

Boxes

- Open the dropdown menu ‘Addins’ for the RStudio toolbar
- Look for the INBOmd package[5]
- Choose insert ‘block’, insert ‘alertblock’ or insert ‘exampleblock’
- Boxes can have an optional title
  - Works only with pdf output!

[5] This is an example of a reusable footnote.

Adding introduction chapters

- Add `{-}` at the end of the header to make it an unnumbered header
- For pdf only
  - Place `\mainmatter` at the position where the table of contents should go
  - The default language is Dutch
    - Place English text between `\benglish` and `\eenglish`
    - Place French text between `\bfrench` and `\efrench`

```
# Voorwoord {-}

\benglish

# English abstract {-}

\eenglish

\mainmatter
```

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Adding appendices

For pdf only

- Insert `\appendix`
- The bibliography will be placed before the appendices
- Chapters in the appendix get letters instead of numbers

Other output formats ignore `\appendix`

- The bibliography will be placed at the end of the document
- Appendix chapters will get a continuing number

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Using templates within a document

`knit_expand` concept

- Create a template with template tags
- Create Rmarkdown through find and replace the template tags with a value
- Repeat for each value
- Combine and knit the resulting Rmarkdown

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Creating the template

- Start by creating a regular Rmarkdown subdocument
- Use a name starting with underscore (_)
- Create a template tag by placing it between double curly brackets `{{` and `}}`.
- Pro tips:
  - Use a single template tag
  - Use that single template tag to select relevant data
  - Add the template tag to the chunk to create unique chunk names

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Example template

```
```{r mt-{{id}}-setup}
if (interactive()) {
  this_id <- sample(mt$id, 1)
} else {
  this_id <- "{{id}}"
}
selection <- filter(mt, id == this_id)
```

`r selection$title[1]`

```{r mt-{{id}}-plot, fig.cap = selection$caption[1]}
ggplot(selection, aes(x = year, y = value)) + geom_point()
```
```

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Handle the template

- Chunk option `results` must be “asis”
- Call `knit_expand()` on each value of the template tag
  - The example will replace every `{{id}}` from the template with a value from `to_do`
- `knit_expand()` only does the ‘find and replace’
- `knit()` actually runs the Rmarkdown code created by `knit_expand()`

```
```{r display-monthly-totals, results = "asis"}
to_do <- unique(mt$id)
rmd <- sapply(
  to_do,
  function(id) {
    knit_expand("_monthly_totals.Rmd", id = id)
  }
)
rmd <- paste(collapse = "\n\n")
cat(knit(text = rmd, quiet = TRUE))
```
```

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Referenties I

Agresti, A. (2002). Categorical Data Analysis. John Wiley and Sons, Hoboken, New Jersey. 710 p.

Amano, T. (2012). Unravelling the dynamics of organisms in a changing world using ecological modelling. Ecological Research 27 (3): 495–507. DOI: https://doi.org/10.1007/s11284-012-0928-6.

Anselin, A., Devos, K. & Vermeersch, G. (2014). Blauwdruk Vogels. In: De Knijf G., Westra T., Onkelinx T., Quataert P. & Pollet M. (eds.), Monitoring Natura 2000-soorten en overige soorten prioritair voor het Vlaams beleid, Rapporten van het Instituut voor Natuur- en Bosonderzoek 2011, p. 188–211. Instituut voor Natuur- en Bosonderzoek, Brussels, Belgium.

Banerjee, S., Carlin, B.P. & Gelfand, A.E. (2003). Hierarchical Modeling and Analysis for Spatial Data. Chapman and Hall, Boca Raton, Florida, USA.